

1. Creating and Renaming Files/Directories

Q - Create a directory named test_dir using mkdir.

Ans - Command [mkdir] - Purpose [to create a directory (folder)]

```
sandeep@sandeep-desktop: /tutedude/assignment1
sandeep@sandeep-desktop:/tutedude/assignment1$ sudo mkdir test_dir
sandeep@sandeep-desktop:/tutedude/assignment1$ ls
test_dir
sandeep@sandeep-desktop:/tutedude/assignment1$
```

Q - Inside test_dir, create an empty file called example.txt.

Ans - using touch command to create a file named - example.txt

```
sandeep@sandeep-desktop: /tutedude/assignment1/test_dir
sandeep@sandeep-desktop:/tutedude/assignment1/test_dir$ sudo touch example.txt
[sudo] password for sandeep:
sandeep@sandeep-desktop:/tutedude/assignment1/test_dir$ ls
example.txt
sandeep@sandeep-desktop:/tutedude/assignment1/test_dir$
```

Q - Rename example.txt to renamed_example.txt using mv

Ans - using mv command i changed the name of txt file.

```
sandeep@sandeep-desktop:/tutedude/assignment1/test_dir$ sudo mv example.txt renamed_example.txt
sandeep@sandeep-desktop:/tutedude/assignment1/test_dir$ ls
renamed_example.txt
sandeep@sandeep-desktop:/tutedude/assignment1/test_dir$
```

2. Viewing File Contents

Q - Use cat to display the contents of /etc/passwd.

Ans - using cat command i showed data of passwd file.

```
sandeep@sandeep-desktop: /etc
sandeep@sandeep-desktop:/etc$ cat passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
_apt:x:42:65534:/:nonexistent:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:998:998:systemd Network Management:/:/usr/sbin/nologin
systemd-timesync:x:996:996:systemd Time Synchronization:/:/usr/sbin/nologin
dhcpcd:x:100:65534:DHCP Client Daemon,,,:/usr/lib/dhcpcd:/bin/false
messagebus:x:101:101:/:nonexistent:/usr/sbin/nologin
syslog:x:102:102:/:nonexistent:/usr/sbin/nologin
```

Q - Display only the first 5 lines of /etc/passwd using head.

Ans - using head command i showed few upper lines of passwd file.

```
sandeep@sandeep-desktop:/etc$ head passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
sandeep@sandeep-desktop:/etc$
```

Q - Display only the last 5 lines of /etc/passwd using tail.

Ans - using tail command i showed few lower lines of passwd file.

```
sandeep@sandeep-desktop:/etc$ tail passwd
tcpdump:x:114:119::/nonexistent:/usr/sbin/nologin
speech-dispatcher:x:115:29:Speech Dispatcher,,,:/run/speech-dispatcher:/bin/false
fwupd-refresh:x:988:988:Firmware update daemon:/var/lib/fwupd:/usr/sbin/nologin
geoclue:x:116:120::/var/lib/geoclue:/usr/sbin/nologin
cups-browsed:x:117:105::/nonexistent:/usr/sbin/nologin
saned:x:118:123::/var/lib/saned:/usr/sbin/nologin
hplip:x:119:7:HPLIP system user,,,:/run/hplip:/bin/false
colord:x:120:124:colord colour management daemon,,,:/var/lib/colord:/usr/sbin/nologin
sandeep:x:1000:1000:Sandeep Singh,,,:/home/sandeep:/bin/bash
sssd:x:121:126:SSSD system user,,,:/var/lib/sss:/usr/sbin/nologin
sandeep@sandeep-desktop:/etc$
```

3. Searching for Patterns

Q - Use grep to find all lines containing the word "root" in /etc/passwd.

Ans - i use grep -n command to show matching results of 'root' in passwd file, the -n flag show the line no where result found in the file.

```
sandeep@sandeep-desktop: /etc
sandeep@sandeep-desktop:/etc$ grep -n 'root' passwd
1:root:x:0:0:root:/root:/bin/bash
37:nm-openvpn:x:112:116:NetworkManager OpenVPN,,,:/var/lib/openvpn/chroot:/usr/sbin/nologin
sandeep@sandeep-desktop:/etc$
```

4. Zipping and Unzipping

Q - Compress the test_dir directory into a file named test_dir.zip using zip.

Ans - using zip -r command i zipped the directory test_dir into test_dir.zip

```
sandeep@sandeep-desktop: /tutedude/assignment1
sandeep@sandeep-desktop:/tutedude/assignment1$ sudo zip -r test_dir.zip test_dir
adding: test_dir/ (stored 0%)
adding: test_dir/renamed_example.txt (stored 0%)
sandeep@sandeep-desktop:/tutedude/assignment1$ ls
test_dir  test_dir.zip
sandeep@sandeep-desktop:/tutedude/assignment1$
```

Q - Unzip test_dir.zip into a new directory named unzipped_dir.

Ans - using unzip command i unzipped the test_dir.zip file

```
sandeep@sandeep-desktop: /tutedude/assignment1
sandeep@sandeep-desktop:/tutedude/assignment1$ sudo unzip test_dir.zip
Archive:  test_dir.zip
replace test_dir/renamed_example.txt? [y]es, [n]o, [A]ll, [N]one, [r]ename: r
new name: test_dir_zip
extracting: test_dir_zip
sandeep@sandeep-desktop:/tutedude/assignment1$ ls
test_dir  test_dir.zip  test_dir_zip
```

5. Downloading Files

Q - Use wget to download a file from a URL (e.g., <https://example.com/sample.txt>).

Ans - I downloaded simple.txt file from specific url -

(<https://sample-files.com/downloads/documents/txt/simple.txt>)

```
sandeep@sandeep-desktop: /tutedude/assignment1
sandeep@sandeep-desktop:/tutedude/assignment1$ sudo wget https://sample-files.com/downloads/documents/txt/simple.txt
[sudo] password for sandeep:
--2026-07-17 11:52:33-- https://sample-files.com/downloads/documents/txt/simple.txt
Resolving sample-files.com (sample-files.com)... 15.235.219.232
Connecting to sample-files.com (sample-files.com)|15.235.219.232|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 225 [text/plain]
Saving to: 'simple.txt'

simple.txt          100%[=====>]          225  --.-KB/s    in 0s

2026-07-17 11:52:33 (72.4 MB/s) - 'simple.txt' saved [225/225]

sandeep@sandeep-desktop:/tutedude/assignment1$ ls
simple.txt  test_dir  test_dir.zip  test_dir_zip
sandeep@sandeep-desktop:/tutedude/assignment1$
```

6. Changing Permissions

Q - Create a file named secure.txt and change its permissions to read-only for everyone using chmod.

Ans - first i creat a file using touch command, then i show file permissions using ls -l , Then i modify permissions of file using chmod 444 secure.txt

```
sandeep@sandeep-desktop: /tutedude/assignment1
sandeep@sandeep-desktop:/tutedude/assignment1$ sudo touch secure.txt
sandeep@sandeep-desktop:/tutedude/assignment1$ ls
secure.txt  simple.txt  test_dir  test_dir.zip  test_dir_zip
sandeep@sandeep-desktop:/tutedude/assignment1$ ls -l secure.txt
-rw-r--r-- 1 root root 0 Jul 17 12:08 secure.txt
sandeep@sandeep-desktop:/tutedude/assignment1$ sudo chmod 444 secure.txt
sandeep@sandeep-desktop:/tutedude/assignment1$ ls -l secure.txt
-r--r--r-- 1 root root 0 Jul 17 12:08 secure.txt
sandeep@sandeep-desktop:/tutedude/assignment1$
```

7. Working with Environment Variables

Q - Use export to set a new environment variable called MY_VAR with the value "Hello, Linux!".

Ans - I used export command to set a variable and print the value of variable using echo command.

```
sandeep@sandeep-desktop: /tutedude/assignment1
sandeep@sandeep-desktop: /tutedude/assignment1$ export MY_VAR="Hello, Linux!"
sandeep@sandeep-desktop: /tutedude/assignment1$ echo $MY_VAR
Hello, Linux!
sandeep@sandeep-desktop: /tutedude/assignment1$
```